

Feedback — Module 2

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You submitted this quiz on **Sat 18 Jul 2015 12:55 PM PDT**. You got a score of **9.00** out of **9.00**.

Question 1

Boyer-Moore: How many alignments are skipped by the good suffix rule in this scenario?

T: GGCTATAATGCGTA

P: TAATTAA

You entered:

3

Your Answer		Score	Explanation
3	✓	1.00	
Total		1.00 / 1.00	

Question 2

Boyer-Moore, true or false: for given P and T, it's possible that some characters from T will never be examined, i.e., won't be involved in any character comparisons.

Your Answer		Score	Explanation
☒ True	✓	1.00	
☐ False			
Total		1.00 / 1.00	

Question 3

Consider a version of Boyer-Moore that uses only the bad character rule (no good suffix rule), and say our pattern P is a random string of 50% As and 50% Ts. In which scenario would you expect Boyer-Moore to skip the most alignments?

Your Answer	Score	Explanation
☐ The text T consists of 40% As, 40% Ts, 10% Cs and 10%Gs		
☒ The text T consists of 10% As, 10% Ts, 40% Cs and 40%Gs	✓ 1.00	
☐ The text T consists of 25% As, 25% Ts, 25% Cs and 25%Gs		
Total	1.00 / 1.00	

Question 4

The naive exact matching algorithm preprocesses:

Your Answer	Score	Explanation
☐ The pattern P		
☐ Both		
☐ The text T		
☒ Neither	✓ 1.00	
Total	1.00 / 1.00	

Question 5

The Boyer-Moore algorithm preprocesses:

Your Answer	Score	Explanation
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- | | | |
|--|---|------|
| ☒ The pattern P | ✓ | 1.00 |
| ☐ The text T | | |
| ☐ Neither | | |
| ☐ Both | | |

Total	1.00 / 1.00
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Question 6

In which of the these scenarios is an offline matching algorithm not appropriate?

Your Answer	Score	Explanation
☒ Your web browser's "find" function that allows you to find a particular word on the web page you are currently viewing	✓ 1.00	
☐ A tool that searches for words in an archive of every speech made in the U.S. Congress		
☐ A tool that evaluates a password by comparing it against a large database of bad (easy-to-guess) passwords		
Total	1.00 / 1.00	

Question 7

Say we have a k-mer index containing all 5-mers from T. We query the index using the first 5-mer from P and the index returns a single index hit. What can we say about whether P occurs in T? Assume T is longer than P and that P is at least 6 bases long.

Your Answer	Score	Explanation
☐ It definitely does		
☐ It definitely does not		
☒ We don't know; not enough information	✓ 1.00	
Total	1.00 / 1.00	

Question 8

Say we have a k-mer index containing all k-mers from T and we query it with 3 different k-mers from the pattern P. The first query returns 0 hits, the second returns 1 hit, and the third returns 3 hits. What can we say about whether P occurs in T?

Your Answer	Score	Explanation
☐ It definitely does		
☒ It definitely does not	✓ 1.00	
☐ We don't know; not enough information		
Total	1.00 / 1.00	

Question 9

Which of the following is not an "edit" allowed in edit distance:

Your Answer	Score	Explanation
☒ Transposition	✓ 1.00	
☐ Deletion		
☐ Substitution		
☐ Insertion		
Total	1.00 / 1.00	